

1. Compute the Taylor polynomial of degree 4 for $f(x) = \cos(2x)$ around basepoint 0.

2. Calculate the velocity, speed and acceleration of a point moving along the curve

$$f(t) = (\exp(t) \cos(t), \exp(t) \sin(t)).$$

Calculate the length of the curve f in $0 \leq t \leq 3$.

①

around base point 0

1.) Calculate $f(x), f'(x), f''(x), f'''(x), f^{(4)}(x)$
 $f(0), f'(0), f''(0), f'''(0), f^{(4)}(0)$

$$\begin{aligned} f(x) &= \cos(2x) \\ f(0) &= \cos(0) = 1 \end{aligned}$$

$$f'(x) = [\cos(2x)]'$$

2 functions, so I will use
CHAIN RULE $\rightarrow f'(x) = i'(j(x)) \cdot j'(x)$
 $i(x) = \cos(x)$
 $j(x) = 2x$

$$\begin{aligned} f'(x) &= -\sin(2x) \cdot 2 \\ f'(x) &= -2 \sin 2x \\ f'(0) &= -2 \cdot \sin(0) = -2 \cdot 0 = 0 \end{aligned}$$

$$f''(x) = [-2 \sin(2x)]'$$

$$i(x) \cdot j'(x) = i(x) \cdot g'(x) + i'(x) \cdot j(x)$$

$$-2 \cdot [\sin(2x)]' + [-2]' \cdot \sin(2x)$$

$$-2 \cdot [\sin(2x)]' + 0 \cdot \sin(2x) =$$

$$f''(x) = -2 [\sin(2x)]'$$

CHAIN RULE

$$f''(x) = -2 \cos(2x) \cdot 2$$

$$f''(x) = -4 \cos(2x)$$

$$f''(0) = -4 \cos(0) = -4$$

$$f'''(x) = [-4 \cos(2x)]'$$

$$f'''(x) = -4 \cdot [\cos(2x)]'$$

$$f'''(x) = -4 \cdot (-\sin(2x) \cdot 2)$$

$$f'''(x) = 8 \sin(2x)$$

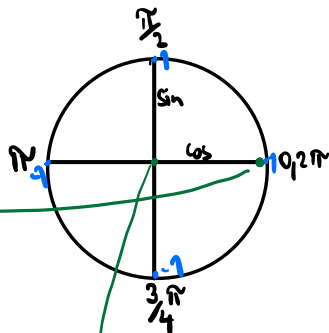
$$f'''(0) = 8 \cdot \sin(0) = 0$$

$$f^{(4)}(x) = [8 \sin(2x)]'$$

$$f^{(4)}(x) = 8 \cdot \cos(2x) \cdot 2$$

$$f^{(4)}(x) = 16 \cdot \cos(2x)$$

$$f^{(4)}(0) = 16 \cdot \cos(0) = 16$$



2.) Use formula: $T(x) = f(a) + \frac{f'(a)}{1!} \cdot (x-a)^1 + \frac{f''(a)}{2!} \cdot (x-a)^2 + \dots$

$$T(x) = f(0) + \frac{f'(0)}{1!} (x-0)^1 + \frac{f''(0)}{2!} (x-0)^2 + \frac{f'''(0)}{3!} (x-0)^3 + \frac{f^{(4)}(0)}{4!} (x-0)^4$$

$$T(x) = 1 + \frac{0}{1} \cdot (x) + \frac{-4}{2} (x)^2 + \frac{0}{6} (x)^3 + \frac{16}{24} (x)^4$$

$$T(x) = 1 - 2x^2 + \frac{2x^4}{3}$$

2. Calculate the velocity, speed and acceleration of a point moving along the curve

$$f(t) = (\exp(t) \cos(t), \exp(t) \sin(t)).$$

Calculate the length of the curve f in $0 \leq t \leq 3$.

$$\begin{aligned} r(\lambda) &= f'(\lambda) \\ a(\lambda) &= f''(\lambda) \\ \text{Speed} &= |r(\lambda)| \end{aligned}$$

② $f(t) = (\exp(t) \cdot \cos(t), \exp(t) \cdot \sin(t))$

$$f'(t) = ([\exp(t) \cdot \cos(t)]', [\exp(t) \cdot \sin(t)]')$$

$$[\exp(t) \cdot \cos(t)]' = \exp(t) \cdot [\cos(t)]' + [\exp(t)]' \cdot \cos(t)$$

CHAIN RULE

$$[f(x) \cdot g(x)]' = f'(x) \cdot g(x) + f(x) \cdot g'(x)$$

$$[\exp(t) \cdot \sin(t)]' = \exp(t) \cdot [\sin(t)]' + [\exp(t)]' \cdot \sin(t)$$

CHAIN RULE

• $f'(t) = (\exp(t) \cdot (-\sin(t)) + \exp(t) \cdot \cos(t), \exp(t) \cdot \cos(t) + \exp(t) \cdot \sin(t)) = v(t) \rightarrow \text{Velocity}$

$$f''(t) = ([\exp(t) \cdot (-\sin(t)) + \exp(t) \cdot \cos(t)]', [\exp(t) \cdot \cos(t) + \exp(t) \cdot \sin(t)]')$$

$$[\exp(t) \cdot (-\sin(t)) + \exp(t) \cdot \cos(t)]' = \exp(t) \cdot (-\cos(t)) + \exp(t) \cdot (-\sin(t)) + \exp(t) \cdot (-\sin(t)) + \exp(t) \cdot \cos(t)$$

$$[\exp(t) \cdot \cos(t) + \exp(t) \cdot \sin(t)]' = \exp(t) \cdot (-\sin(t)) + \exp(t) \cdot \cos(t) + \exp(t) \cdot \cos(t) + \exp(t) \cdot \sin(t)$$

• $f''(t) = (2(\exp(t) \cdot (-\sin(t))), 2(\exp(t) \cdot \cos(t))) = a(t) \rightarrow \text{ACCELERATION}$

$$\begin{aligned} \text{Speed}(t) &= |v(t)| \\ &= \sqrt{(\exp(t) \cdot (-\sin(t)) + \exp(t) \cdot \cos(t))^2 + (\exp(t) \cdot \cos(t) + \exp(t) \cdot \sin(t))^2} \\ &= \sqrt{\exp^2(t) \cdot (-\sin(t) + \cos(t))^2 + \exp^2(t) \cdot (\cos(t) + \sin(t))^2} \\ &= \sqrt{\exp^2(t) \cdot ((-\sin(t))^2 + (\cos(t))^2) + \exp^2(t) \cdot ((\cos(t))^2 + (\sin(t))^2)} \end{aligned}$$

$$\sin^2(t) + \cos^2(t) = 1$$

$$\cos^2(t) + \sin^2(t) = 1$$

• $\text{speed}(t) = \sqrt{2 \exp^2(t)} \rightarrow \text{MAGNITUDE}$

$$\text{Length} = \int_a^b |f'(t)| dt$$

• $L = \int_0^3 \sqrt{2 \exp^2(t)} = \int_0^3 \sqrt{2} \cdot \exp(t) = \sqrt{2} \cdot \int_0^3 \exp(t) = \sqrt{2} \cdot \exp(t) \Big|_0^3 = \sqrt{2} \cdot \exp(3) - \sqrt{2} \cdot \exp(0) = \sqrt{2} \cdot 20.09 - \sqrt{2} \cdot 1 = \sqrt{2} \cdot (20.09 - 1) = \sqrt{2} \cdot 19.09$

≈ 26.99 LENGTH of CURVE FROM 0 to 3